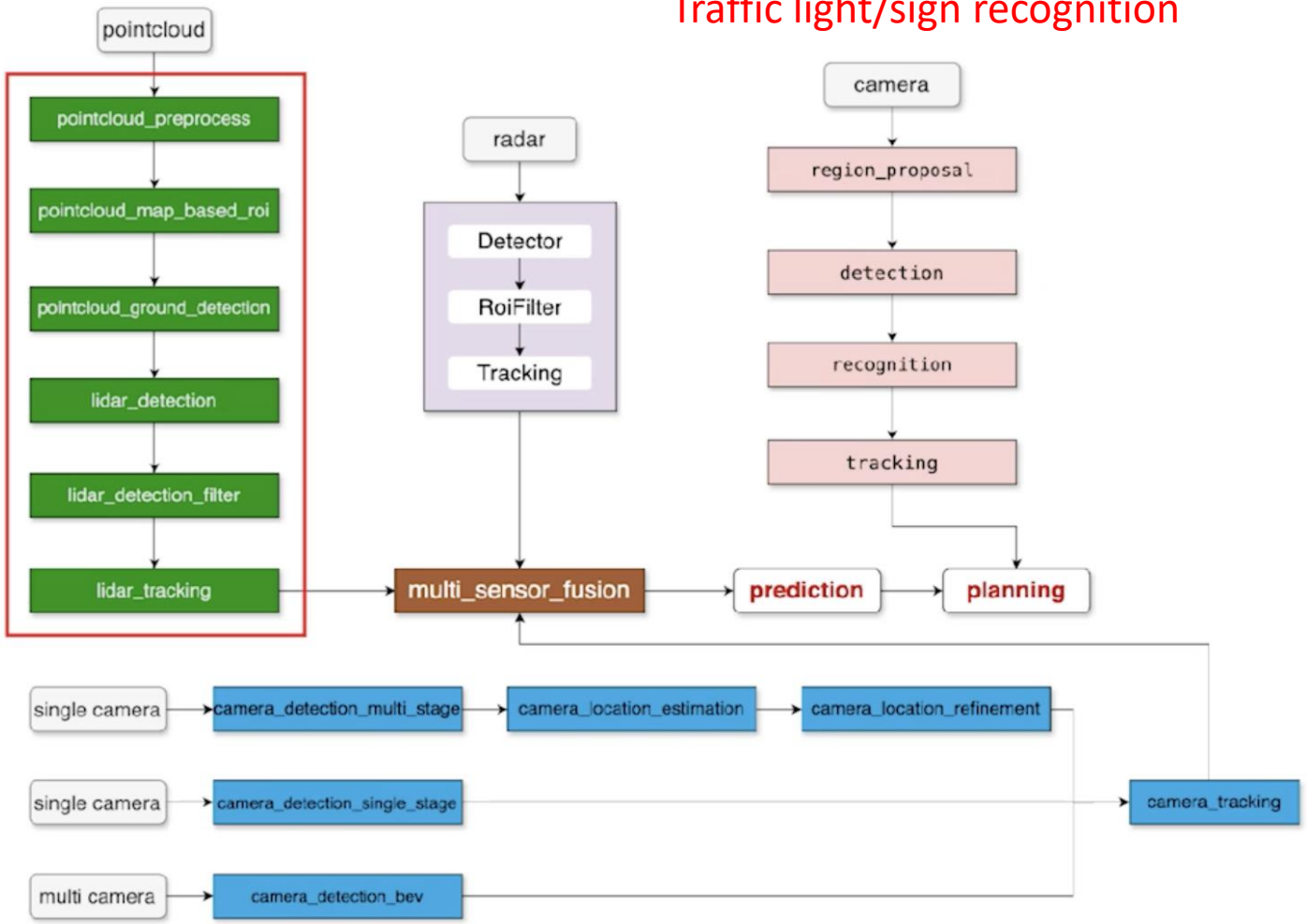
CS5493 Topics in Autonomous Driving

香港城市大學
City University of Hong Kong
Innovating into the Future

Tutorial 6: Perception

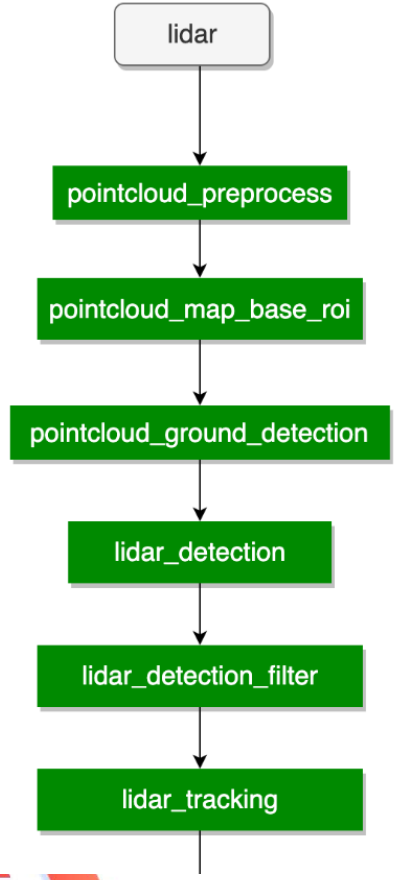
Perception in Apollo

Traffic light/sign recognition



Perception in Apollo

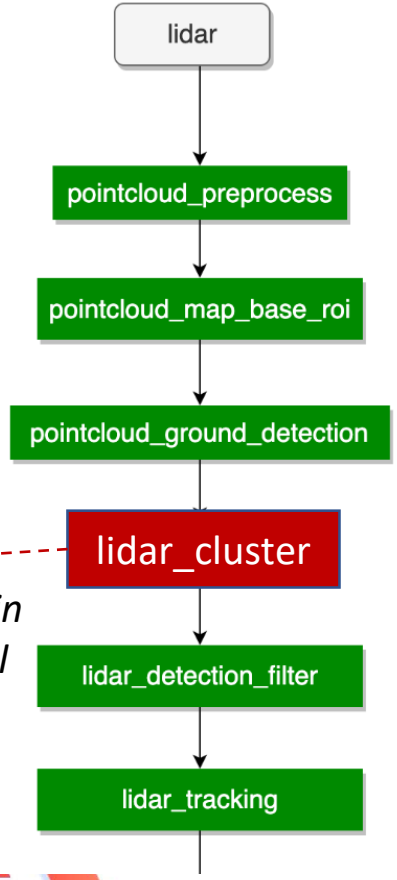
Lidar-based perception



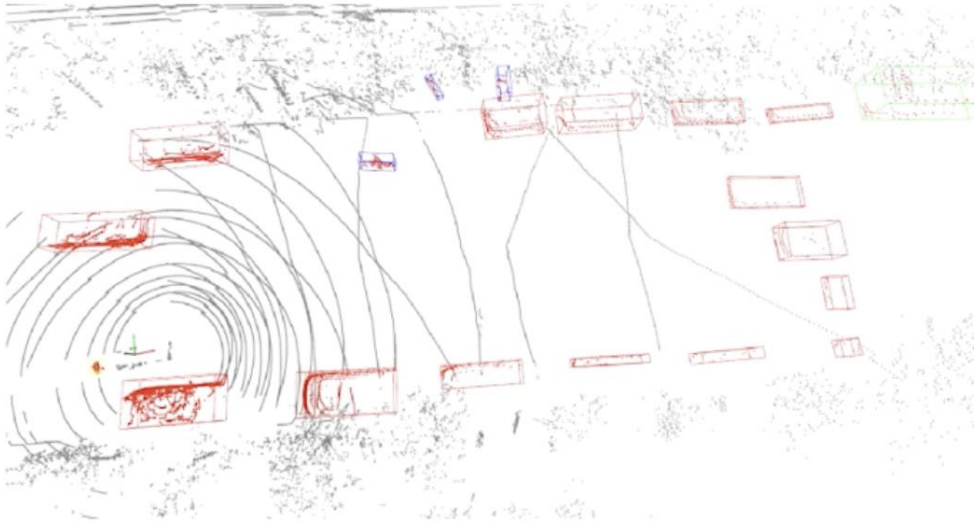
Component	Function
pointcloud_preprocess	Delete outliers, points from ego-vehicle, poin ts that are too far
pointcloud_map_based_roi	Filter pointcloud outside RoI
pointcloud_ground_detection	Filter all points that are considered ground
lidar_detection_filter	Filter items according to rules set before (item categories, …)
lidar_tracking	Track the movement trajectory of obstacles, update the motion state and geometric shap e of the obstacles, and assign tracking IDs.

Perception in Apollo

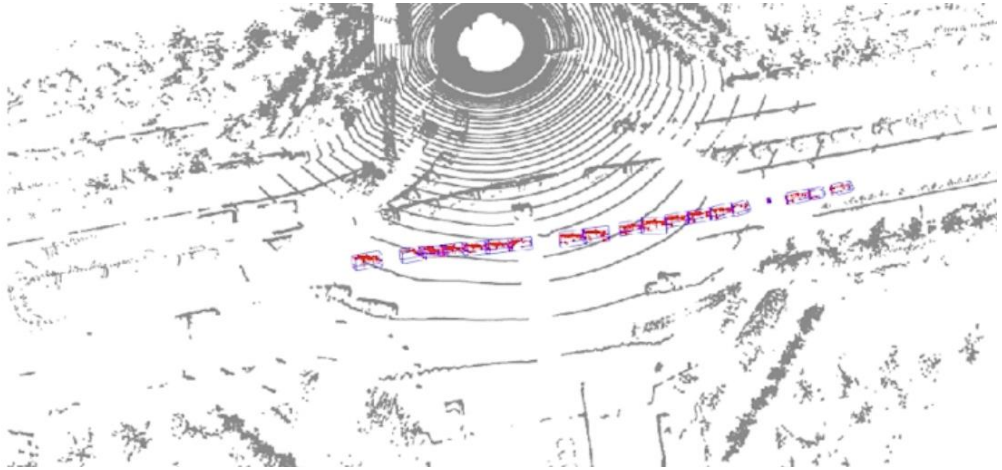
Lidar-based perception



Use this simple algorithm in our tutorial



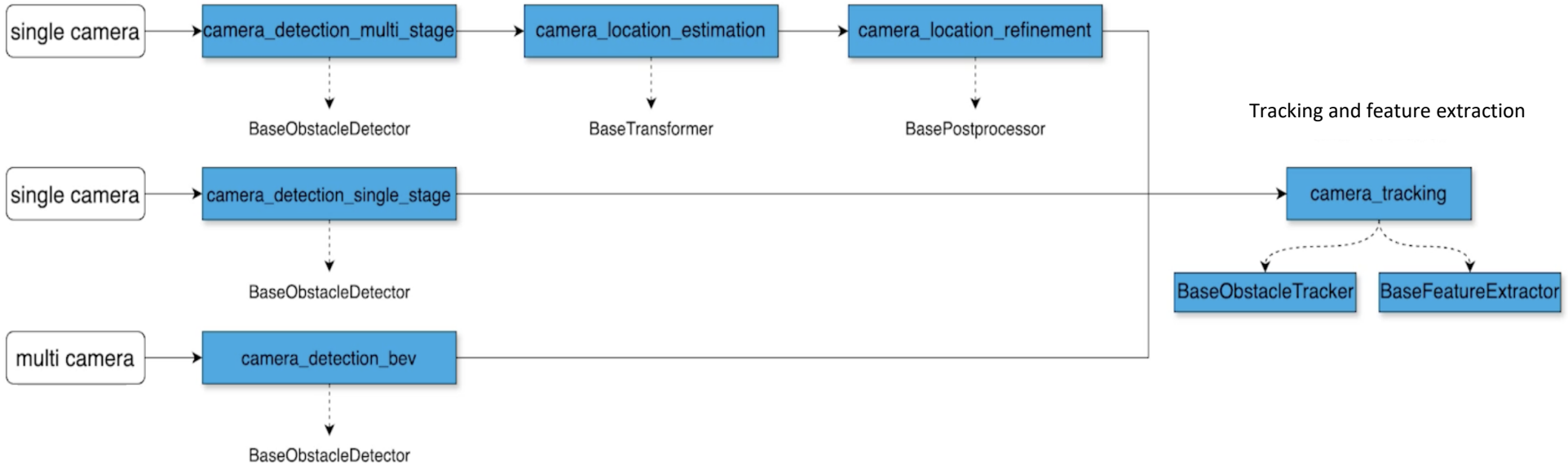
Detection result



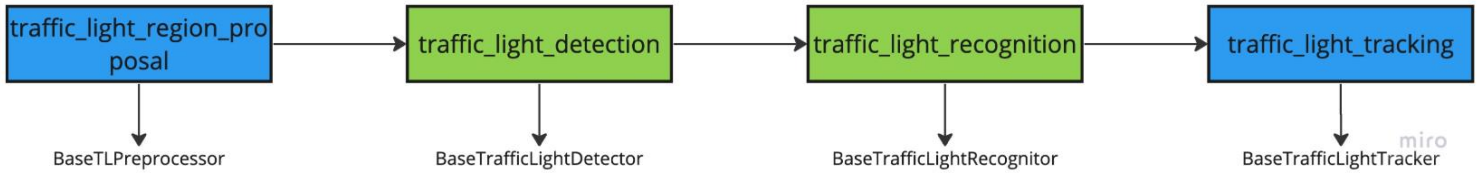
Tracking result

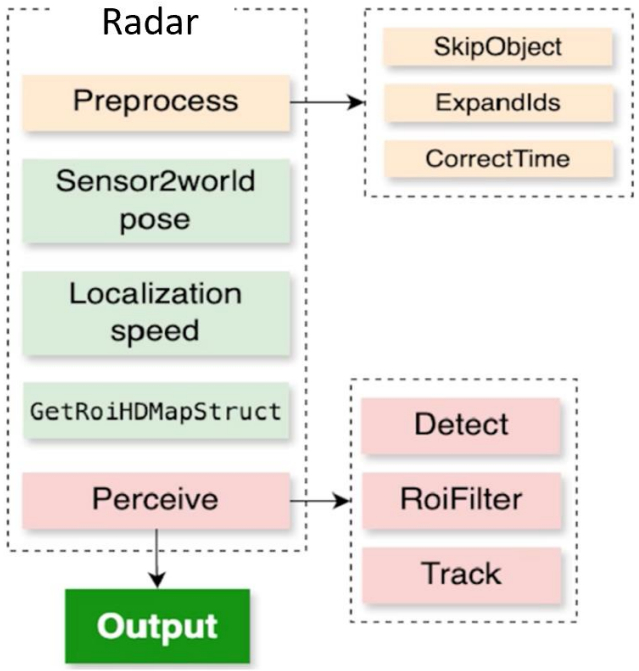
Perception in Apollo

Camera-based perception



Traffic light recognition using camera



Perception in Apollo

Radar-based perception